

GRADE 9 MATHEMATICS · ALGEBRA

Linear Relations

THE BIG IDEA

A linear relation is a straight line with a constant rate of change — **y-intercept and **y-value**.**

THE TWO FORMULAS THAT RUN THIS UNIT

$$m = (y_2 - y_1) / (x_2 - x_1)$$

slope from two points

ions

nt line. **Slope** tells you the
ypt tells you the starting

$m = \text{slope} \cdot b = \text{y-intercept}$

WORKED EXAMPLE

Two points: (1, 3) and (4, 12)

Slope: $m = (12 - 3) / (4 - 1) = 9 / 3 = 3$

Use $y = mx + b$: $3 = 3(1) + b \rightarrow b = 0$

Equation: $y = 3x$ (direct variation — passes through origin)

⚠ WATCH OUT FOR

MISTAKE

“y-intercept is where the line crosses the x-axis”

ses through the origin)

FIX

It crosses the **y-axis**, at $x = 0$

MISTAKE

“ $y = 3x$ is partial variation”



SELF-CHECK

Given $y = -4x + 7$, can you state the slope, y -intercept, and real-world context?

FIX

No constant term means **direct** variation

-intercept, and what both mean in a real